

THE FUTURE OF STATISTICAL INFERENCE IN THE CLASSROOM: ALIGNING THEORY WITH COMPUTATION

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Having students master the details of statistical inference—point and interval estimation, and hypothesis testing—is a standard learning goal in tertiary statistics programs. But in an age where the use of computers has superseded the use of statistical tables, there arises the challenge of determining what details we should teach, and how. In this work, we present approaches to inference that we use in our mathematical statistics classes, where we replace methods employing large-sample approximations, transformations, and statistical tables, etc., with ones that utilize numerical approaches. To be clear, we do not “replace theory with computation” but rather “align theory with computation.” However, we would argue that with proper scaffolding one can incorporate the numerical methods we present into less-theoretical introductory statistics and data science classes.

INTRODUCTION

In tertiary education, the term “mathematical statistics” denotes a calculus-based course sequence that covers concepts of probability and statistical inference, generally over the course of an academic year. This course sequence is typically targeted to those pursuing a degree in statistics and/or data science. The concepts covered within it—elementary probability theory, random variables and univariate and multivariate probability distributions, statistics and sampling distributions, point and interval estimation, hypothesis testing, and statistical modeling—are largely if not entirely the same now as those that would have been covered in a mid-20th century instantiation of the sequence. Panelists at the JSM session *Is the Math Stat Course Obsolete?* (American Statistical Association 2003) referred to these concepts collectively as the “Hogg and Craig syllabus” (while specifically referencing the 1965 second edition of Hogg and Craig’s *Introduction to Mathematical Statistics*). In our opinion, mathematical statistics is *not* obsolete: knowledge of probability distributions and how they are used to make inferences about populations is fundamental to the enterprise of statistics! However, aspects of the Hogg and Craig syllabus that still permeate mass-market textbooks, such as those that rely on the use of large-sample approximations and statistical tables, *have* become obsolete and need to change.

In this paper, we describe how we are modernizing the mathematical statistics course sequence targeted to statistics and data science majors at Carnegie Mellon University, to make fuller use of computation and to root out “historical inefficiencies.” To be clear, this does not mean replacing attention to theoretical detail—having students learn “formulas and derivations”—with computationally based black-box approaches to problem solving. Rather, it means determining *what* theoretical details are still relevant to impart to students now that the computer has superseded the table as a primary problem-solving tool. Rooting out inefficiencies goes far beyond simply replacing a z-table with R’s `pnorm()` function, as analytical workflows that are implicitly designed with tables in mind pervade the Hogg and Craig syllabus. Below, we identify some of these and show how they may be reformulated to make effective use of numerical root-finding, simulation, and optimization algorithms.

Our work conforms with modern curricular guidelines. For instance, the 2014 American Statistical Association *Curriculum Guidelines for Undergraduate Programs in Statistical Science* (Horton et al. 2014) suggests that “[t]o be prepared for statistics and data science careers, students need...the ability to perform algorithmic problem solving” and that “[t]hey need to...use a variety of computational approaches to extract meaning from data” as well as to “[d]evelop flexible problem-solving skills.” The *GAISE College Report* (Carver et al. 2016) includes the recommendation that students “[u]se technology to explore concepts and analyze data,” while the 2018 National Academies report on *Data Science for Undergraduates: Opportunities and Options* (National Academies of Sciences, Engineering, and Medicine 2018) stresses the need for students to acquire “data acumen” by building upon computational foundations (here, algorithmic thinking and simulations) as well as statistical foundations (inference).

Having said this, we must make clear that this work is about making fundamental changes to how students *approach* problem solving in mathematical statistics classes, and *not* (necessarily) about improving students' understanding of statistical concepts, which is a separate but obviously important issue. As a concrete example, below we show how instead of working with the traditional pivotal method, students can construct confidence intervals by working directly with the sampling distribution of any statistic, not just one that is a pivotal quantity, and how they can easily carry out construction by utilizing a numerical root-finder, regardless of whether an analytic solution is obtainable. This greatly broadens the range of solvable interval estimation problems while at the same time making interval estimation more accessible to those with less mathematical training. However, even if one were to adopt this approach, *one would still need to teach students what confidence intervals represent and how to properly interpret them*. Our need as teachers to foster conceptual understanding is unchanged, as is how we attempt to build that understanding, such as by utilizing methods of simulation-based inference (à la Rossman & Chance 2014). In the end, this work is not about changing A (collecting and understanding data) nor about changing B (using acquired conceptual understanding to properly interpret analysis results); it is about offering modern algorithmic approaches for getting from A to B.

TRANSFORMATIONS

Statistical inference that revolves around making inferences about the population mean μ and/or variance σ^2 of a normal distribution, given n independent and identically distributed (iid) data, is a standard topic in mathematical statistics. In traditional settings, we utilize three separate variable transformations that depend on our inferential goal and our knowledge of the population variance:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1) \quad T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1) \quad W = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1),$$

where $\bar{X}$ and S are the sample mean and standard deviation, respectively. While students cannot avoid the use of the quantity T , which depends on two random variables, they need not work with either Z or W , as these exist primarily to “reformat” numerical exercises so that they may be solved using statistical tables. Students should work with the following sampling distributions instead:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \quad S^2 \sim \text{Gamma}\left(\frac{n-1}{2}, \frac{2\sigma^2}{n-1}\right).$$

Working with these distributions is more straightforward for students, as they can get from mathematical expressions to, e.g., solution-generating R codes that much more easily. However, while students need not work directly with the standard normal and chi-square distributions in analyses, we should still cover their details so that our students can understand how others still utilize them in practice.

But, regarding the use of T : a long-term issue with mass-market textbooks is their presentation of an entirely false dichotomy between “small-sample” and “large-sample” approaches to inference that utilize t and z tables, respectively. This dichotomy *only exists* because t tables are limited to small sample sizes (typically $n \leq 30$), and its effect is to sow confusion. We must make sure that when students are given iid data sampled according to a normal distribution of unknown mean and variance, they *always* utilize the t distribution via computer functions like R's `pt()`, regardless of the sample size.

NUMERICAL ROOT-FINDING

The three random variables discussed above— Z , T , and W —are “pivotal quantities”: they are all functions of both statistics and parameters of interest but have sampling distributions that are not functions of the parameters of interest. Having students define pivotal quantities is central to traditional approaches to generating interval estimates, but as we show below, it is unnecessary for them to do so.

Freeman (2026b) describes how one can estimate bounds for random intervals by solving

$$F_Y(y_{\text{obs}}|\theta) - q = 0$$

for the sampling distribution parameter θ . We dub this method “cdf inversion.” To be clear, we did not develop this method—it is described in, for instance, Casella & Berger (2002)—but it is underutilized in mathematical statistics settings. Here, Y denotes a statistic whose cumulative distribution function is F_Y , while y_{obs} is the observed value of the statistic and q is a constant that is related to the confidence coefficient $1 - \alpha$. We provide students with tables that relate α to q in different situations; for instance, when computing a lower bound of a two-sided 95% confidence interval for the normal population mean μ , q is set equal to $1 - \alpha/2$, or 0.975. Key to cdf inversion is that the statistic Y *need not be a pivotal*

quantity! (It can be, but need not be.) This simplifies the interval estimation workflow by eliminating the step of finding a pivot. Students can sometimes solve the equation above by hand, but they can always utilize a numerical root-finder such as R's `uniroot()` function, as we do in the following example where n data are sampled according to a binomial distribution. We determine a one-sided 95% lower bound on the success probability p using the statistic $Y = \sum_{i=1}^n X_i$, for which the sampling distribution is $\text{Binom}(nk, p)$. (Note that Y can be based on any inferentially relevant statistic, such as the sample median; it need not be a sufficient statistic, as it is here.)

```
k      <- 5
x.obs <- c(2, 4, 1, 2, 1) # n=5 replications, with k=5 trials per replication
f <- function(p, k, y.obs, n, q) {
  pbinom(y.obs, size=n*k, prob=p) - q
}
uniroot(f, interval=c(0,1), k=5, y.obs=sum(x.obs)-1, n=5, q=0.95)$root # root: 0.236
```

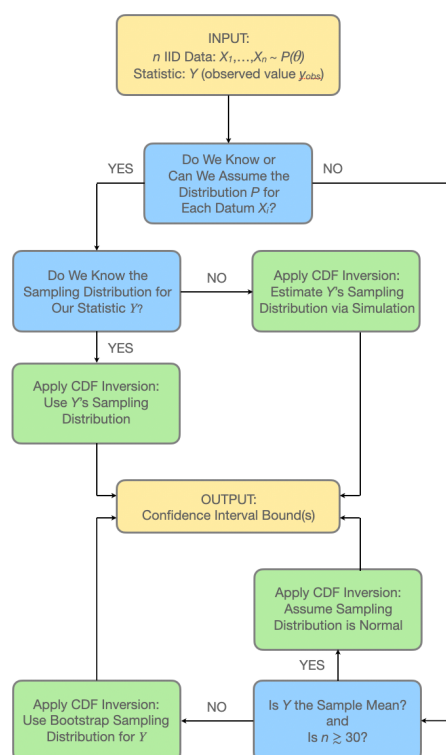
The function `f()` evaluates the equation given above, while `uniroot()` searches for the value of p between 0 and 1 such that `f()` returns zero (within machine tolerance). Note how we subtract 1 from the observed statistic value: this is a “discreteness correction” whose use leads to our computing a correct Clopper-Pearson lower bound. The use of a root-finder greatly expands the range of distributions for which we can compute interval estimates, a fact that students quickly realize: in a Fall 2025 assignment given just after the introduction of both the cdf inversion and pivotal methods, 95% of students were already able to identify cdf inversion as the more applicable method (Freeman 2026b). It is important to stress here that one should not view using a root-finder as approximation: if we know the exact sampling distribution from which the statistic Y is drawn, then any determined bound is exact. We contrast this with, for example, the use of the Central Limit Theorem, where any derived bound will be only approximately correct because the assumed sampling distribution for Y is not the exact one.

An interval estimation flowchart that we provide to students is displayed in Figure 1. A key takeaway from this flowchart is that cdf inversion is applicable in a wide range of analytical situations: when we know the sampling distribution for the statistic Y exactly, or when we know it only approximately (such as when we apply the CLT), or even when we can only estimate it using (non)parametric bootstrapping. But there is another use case as well, as we demonstrate below: if we can assume the distribution according to which each datum is sampled, but we cannot specify the sampling distribution for the statistic, we can combine numerical simulation with root-finding to determine bounds.

We note that we can also utilize the equation given above in hypothesis testing: for instance, we can replace y_{obs} with y_{RR} and θ with the null hypothesis value θ_0 and solve for a rejection-region boundary y_{RR} . This typically does not involve the explicit use of numerical root-finding, as we are simply solving an inverse cdf problem and thus we can use standard functions like R's `qnorm()`. For more information on how we work with sampling distribution cdfs in hypothesis testing, see Freeman (2026a).

While it might initially appear that by using a numerical root-finder we are promoting a “black-box” approach to inference, there is still much underlying theoretical detail that a mathematical statistics student would have to engage with before any code can be written down. A student might need to (a) identify the sampling distribution for the statistic Y , by using, for instance, the method of moment-generating functions, (b) derive the cumulative distribution

Figure 1. A flowchart that students can use to determine the appropriate way to determine an interval estimate for a parameter θ of a specified distribution P , given n iid data and a chosen statistic Y . Ultimately, we can apply cdf inversion to any sampling distribution—exact, approximate, or estimated via simulations or bootstrapping. (Based on Freeman 2026b.)



function for Y , or (c) derive the expected value for Y , to determine the correct value for q . A student might even determine whether, for example, Y is a pivotal quantity or a sufficient statistic, or whether inverting the cdf can be done analytically, etc. *So, while some of the theoretical details might change, the overall level of engagement with them does not.* Having said this, we would argue that properly scaffolded computation-based approaches can make interval estimation more accessible to students who have at least some programming experience and relatively less mathematical training...which is a substantial population in tertiary education. “Properly scaffolded” means that some or all theoretical details are abstracted away and that we encapsulate the computational details within applets or clearly defined recipes. As stated in the 2014 American Statistical Association curriculum guidelines (Horton et al. 2014), “[f]ew undergraduate students need the mathematics used to derive classical statistical formulas, many of which are often superseded by computational approaches.” We call on teachers of introductory statistics and data science to think about how they might implement the interval-estimation approach described above, and the numerically based approaches shown below, in their own classes.

NUMERICAL SIMULATION

An obvious use for computers in mathematical statistics classes is simulation: the generation of large numbers of datasets from specified distributions. Through numerical simulation, we can determine empirical cumulative distribution functions for statistics of our generated data and apply these cdfs in probability calculations, confidence interval estimation, and hypothesis testing. Any inferential results that we derive using empirical cdfs are effectively exact if the distribution of each datum is specified and the number of simulated datasets is large. Simulation is fast—we can generate accurate and precise results on typical computers in ~ 1 -10 CPU seconds—and easy to concisely code.

As a first example, we show how we can combine simulations with root-finding to numerically derive the bounds for a 95% confidence interval for a when we sample n iid data according to $\text{Beta}(a, b)$ distribution, with b fixed. Students might determine that a computationally amenable sufficient statistic for a , whose value increases with a , is $Y = -\sum_{i=1}^n \log X_i$, but they will find that the sampling distribution for Y is unknown. To determine the lower bound on the confidence interval, we would mimic the code shown above, but instead of providing a built-in R function that returns the cdf value, we would insert a simulation codelet that returns the empirical cdf value instead:

```
x.obs <- c(0.3313, 0.1908, 0.1089, 0.0006937, 0.1642)
y.obs <- -sum(log(x.obs))
f <- function(a, n, b, y.obs, q, num.sim=100000, seed=236) {
  set.seed(seed)
  X <- matrix(rbeta(n*num.sim, shape1=a, shape2=b), nrow=num.sim)
  Y <- apply(X, 1, function(x){-sum(log(x))})
  sum(Y <= y.obs)/num.sim - q
}
uniroot(f, c(0.01,2), n=5, b=2.6, y.obs=y.obs, q=0.975)$root # root: 1.030
```

Here, $\text{sum}(Y \leq y.\text{obs})/\text{num.sim}$, the empirical cdf value, has replaced $F_Y(y_{\text{obs}}|a)$ in the cdf-inversion workflow. As noted above, we can plug any statistic into this codelet, not just a sufficient statistic: for instance, if we want to infer the value of the normal mean μ using just the second-largest datum in a dataset (if for no other reason than to have students to engage with order statistics), we can do that.

As a second example, we show how we can use simulation-based empirical cdfs to compute a hypothesis test p -value, and in so doing avoid the use of a large-sample approximation (Freeman 2025). Let’s suppose that we have carried out a set of k multinomial trials, each with m possible outcomes, and we wish to test the null hypothesis $p = \{p_1 = p_{1,o}, \dots, p_m = p_{m,o}\}$. In such a situation, the use of Pearson’s chi-square goodness-of-fit (GoF) test is ubiquitous. However, the assumption that the chi-square statistic is sampled according to a chi-square probability density function is a large-sample approximation, based on the fact that a multinomial random variable asymptotically converges in distribution to a multivariate normal random variable. As the number of expected counts associated with each outcome goes to zero, p -values generated by the GoF test become increasingly biased, with the true Type I error rate α becomes larger than expected. (This leads to the convention that we should not use the GoF test if the number of expected counts for any outcome, $kp_{i,o}$, is five or less.) When we

simulate data from the multinomial distribution under the null, there are no expected-counts constraints, and because the sampling distribution is exact, the p -value estimate—namely, the proportion of simulated probability mass function values that are equal to or less than the pmf value we observe—is unbiased. Let's assume that we toss a six-sided die $k = 24$ times...how would we determine whether the die is fair?

```
set.seed(36236)
X.obs <- c(1, 5, 4, 4, 6, 4)          # observed counts for each die face
m      <- length(X.obs)
p.o    <- rep(1/m, m)
num.sim <- 100000
k       <- sum(X.obs)
y.obs   <- dmultinom(X.obs, prob = p.o) # the pmf value for the observed data
X       <- rmultinom(num.sim, k, p.o)
a       <- apply(X, 2, function(x, p){dmultinom(x, prob = p)}, p = p.o)
sum(a <= y.obs)/num.sim              # the estimated p-value

f <- function(p, k, y.obs, q) {      # generate 95% CI for true p value
  pbinom(y.obs, size = k, prob = p) - q
}
y.obs <- sum(a <= y.obs)              # how many "successes" did we observe?
uniroot(f, interval = c(0,1), k = num.sim, y.obs = y.obs-1, q = 0.975)$root
uniroot(f, interval = c(0,1), k = num.sim, y.obs = y.obs, q = 0.025)$root
```

The estimated p -value is 0.5634, with 95% confidence interval [0.5603,0.5665]. Now, it is not the case that we can always forgo the use of large-sample approximations—for instance, we still need to assume that the appropriate sampling distribution in many contingency table-based hypothesis tests is the chi-square distribution—but when we can forgo their use, we should!

NUMERICAL OPTIMIZATION

Numerical optimization arises naturally in different areas of statistical inference, such as logistic regression, but its use is generally “under the hood.” However, explicit numerical optimization is useful, for example, when we perform maximum likelihood estimation (MLE). In traditional mathematical statistics settings, students learn about three different point estimators: the MLE, the minimum-variance unbiased estimator (MVUE), and the method-of-moments estimator (MoM). A simple logic for choosing between these estimators is as follows: if the MLE is analytically derivable, via differentiation of the likelihood function, and it is unbiased, use it because of its asymptotic and invariance properties, but if it is biased, we can attempt to derive the MVUE...but if neither the MLE nor the MVUE are analytically determinable, derive the MoM estimator instead. The existence of computers alters this logic: if we cannot derive the MLE or MVUE, we can code the MLE instead using a numerical optimizer. We no longer need to have students learn MoM estimation, except perhaps as a mechanism for generating good initial parameter values to pass to an optimizer, as we illustrate below.

```
set.seed(36236)
X <- rgamma(15, shape=4, scale=2) # shape MLE is not analytically derivable!
f <- function(par, X)
{
  -sum(log(dgamma(X, shape=par[1], scale=par[2]))) # minimize -log(like)
}

mom.a <- mean(X)^2 / (mean(X^2) - mean(X)^2) # 7.794
mom.b <- mean(X) / mom.a                     # 1.158
par <- c(mom.a, mom.b) # initial guesses: shape and scale
optim(par, f, X=X)$par[1] # 8.269
optim(par, f, X=X)$par[2] # 1.092
```

As is the case for numerical root-finding, one should view numerical optimization of the MLE as a counterpart to analytical derivations: using it gives students the ability to solve a wider variety of point estimation problems than analytic derivations alone will allow. And, as is the case for confidence interval construction, the use of numerical methods makes point estimation that much more accessible to students with relatively less mathematical training.

CONCLUSIONS

The mathematical statistics course sequence is not obsolete, as probability and statistical inference is the foundation that all tertiary students build upon to perform, and properly interpret the results of, data analyses. But in the age of computers, aspects of how we teach mathematical statistics— aspects of what we call the Hogg and Craig syllabus—must change. Transformations whose purpose is to “reformat” problems into ones solvable with statistical tables: often no longer needed. Large-sample approximations that “reformat” problems into ones solvable with statistical tables: also often no longer needed. Tables themselves: definitely no longer needed. Numerical approaches to problem solving— utilizing root-finding, simulation, and optimization—allow students to attack a much wider range of problems than the largely exponential- and normal-based ones that pervade the Hogg and Craig syllabus. To be clear: numerical approaches should not supersede analytical derivations (when they can be made), but should complement them: our students should be proficient in both math and coding!

A beneficial by-product of a move towards numerical approaches in mathematical statistics is that it “lowers the barrier to entry”: students with less mathematical training can more easily approach inferential computations. Thus we feel that these approaches, properly scaffolded, with the theoretical details abstracted away and the computational details encased within apps or decomposed into easy-to-follow recipes, can be extensively utilized in introductory statistics or data science courses. In the future, we welcome working with teachers of these courses to make such apps and recipes a reality.

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